

WeldNet: Vision-Based Weld Inspection for Intelligent Manufacturing Systems

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Abstract

Manual weld inspection is often slow, inconsistent, and prone to subjectivity, especially in high-precision manufacturing industries. This paper presents a WeldNet: AI-based weld defect detection system using deep learning and computer vision for real-time industrial inspection. The proposed methodology detects weld defects with high accuracy while being optimised for edge deployment in manufacturing environments. Our approach improves inspection consistency, reduces production delays, and supports faster quality assurance workflows. The system also establishes a foundation for future intelligent welding systems with automated feedback and adaptive process control, contributing towards Industry 4.0 manufacturing.

Keywords: Machine Vision, Weld Inspection, Deep Learning, Industry 4.0, YOLOv8

1 Introduction

Industrial weld inspection is critical for ensuring structural integrity, safety, and product quality in advanced manufacturing systems [Torres-Torres et al., 2026]. However, existing inspection pipelines are still largely manual and rely heavily on human expertise, leading to inconsistent evaluations and limited scalability in high-throughput production environments. To address these limitations, we propose a real-time deep learning-based weld defect detection methodology leveraging computer vision techniques. The system is designed for efficient and accurate detection of weld defects from images and video streams using lightweight object detection models trained on a YOLO-formatted weld dataset. It is robust to real-world variations such as lighting changes, texture diversity, and noise. Beyond defect detection, the framework integrates automated weld logging and intelligent feedback. For each detected defect, the system provides repair suggestions and preventive guidance, enabling both quality control and operator support. Overall, this work contributes an efficient and deployable weld inspection system that improves detection accuracy, reduces manual inspection effort, and enhances traceability for industrial applications aligned with Industry 4.0 requirements.

2 Background

Weld inspection is an important process in industries such as oil and gas, pharmaceuticals, and structural manufacturing, where safety and reliability are critical. Traditionally, weld quality is assessed manually by inspectors. While widely used, this approach is time-consuming and subjective, and it can lead to inconsistent results. Recent advances in artificial intelligence and computer vision have enabled automated defect detection systems that assist and expedite operator-led weld joint inspection. In particular, deep learning-based object detection models have shown strong performance in identifying visual defects in industrial settings when trained and fine-tuned on large annotated datasets. This work develops a deep learning-based weld defect detection system that identifies defects in images and video streams. In addition to detection, the system supports basic reporting

and logging for traceability and quality tracking in manufacturing workflows. The goal is to improve inspection consistency, reduce human error, and support more efficient and reliable industrial weld quality control.

3 Methodology

This work proposes a lightweight vision-based framework for automated weld defect detection in intelligent manufacturing systems. As illustrated in Figure 1, the pipeline consists of image acquisition, preprocessing, data annotation, YOLOv8n-based model training, and evaluation. The goal is accurate defect detection with low computational cost for real-time industrial deployment.

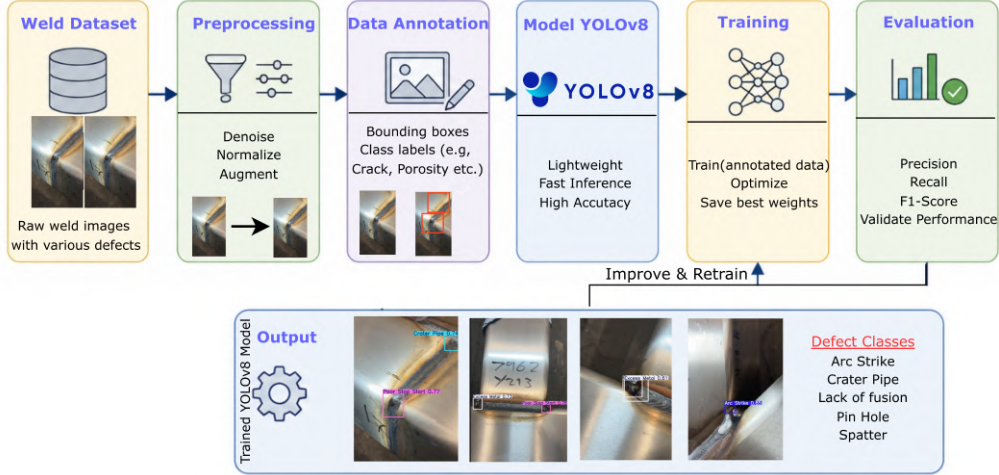


Figure 1: Proposed methodology for automated weld defect detection.

3.1 Data Preprocessing and Augmentation

Weld images containing defects such as arc strike, crater pipe, lack of fusion, pinhole, and spatter are collected during industrial processes. To improve robustness, images are normalized as

$$I_{\text{norm}} = \frac{I - \mu}{\sigma + \epsilon} \quad (1)$$

where μ and σ are dataset mean and standard deviation.

Data augmentation is applied using a stochastic transformation function:

$$I' = \mathcal{T}(I) \quad (2)$$

which includes geometric and photometric variations such as rotation, scaling, flipping, and brightness changes.

3.2 Detection Model

A YOLOv8n-based detector is used due to its efficiency and anchor-free design. The model learns a mapping:

$$f_{\theta}(I) \rightarrow (b_i, c_i, s_i)_{i=1}^N \quad (3)$$

where b_i is the bounding box, c_i is the defect class, and s_i is the confidence score.

Bounding box overlap is measured using Intersection over Union (IoU):

$$\text{IoU}(b, b^*) = \frac{|b \cap b^*|}{|b \cup b^*|} \quad (4)$$

3.3 Training Objective

The model is trained using a multi-task loss:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{box}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{dfl}} \quad (5)$$

where $\mathcal{L}_{\text{box}}$ is based on IoU/CIoU, $\mathcal{L}_{\text{cls}}$ is classification loss, and $\mathcal{L}_{\text{dfl}}$ refines localization.

Performance is evaluated using Precision, Recall, F1-score, and mean Average Precision (mAP). During inference, the trained model predicts defect locations and categories in real time, enabling efficient weld quality inspection in industrial environments.

4 Results

This section evaluates the YOLOv8n-based weld defect detection methodology on a multi-class dataset containing Arc Strike, Crater Pipe, Lack of Fusion, Pin Hole, and Spatter defects. YOLOv8n was selected due to its strong balance between detection accuracy and real-time inference performance. Performance is assessed using Precision, Recall, F1-score, mAP@0.5, and inference speed (FPS).

4.1 Ablation Study

An ablation study is conducted to evaluate the contribution of each component in the proposed pipeline, including normalization, data augmentation, and CIoU-based regression loss. Results in Table 1 show that standard score normalization improves stability under varying illumination, while data augmentation enhances generalization. CIoU loss further improves bounding box accuracy and overall localization performance.

Table 1: Ablation study of key components in the proposed methodology.

Configuration	Precision	Recall	mAP@0.5	F1-score
Baseline (YOLOv8)	88.4	86.1	87.2	87.2
+ Standard Normalization	89.7	87.5	88.6	88.6
+ Data Augmentation	91.2	89.3	90.4	90.2
+ CIoU Loss (Full Model)	93.5	91.8	92.6	92.6

4.2 Comparison with State-of-the-Art Methods

To demonstrate the effectiveness of the proposed method, we compare it with several state-of-the-art object detection models, including Faster R-CNN [Ren et al., 2016], SSD [Liu et al., 2016], YOLOv5 [Jocher et al., 2022], and YOLOv7 [Wang et al., 2023]. All models are evaluated under the same dataset and experimental settings.

As shown in Table 2, the proposed YOLOv8n-based framework achieves the best trade-off between accuracy and inference speed. While two-stage detectors such as Faster R-CNN provide competitive accuracy, they suffer from significantly higher computational cost. In contrast, the proposed method maintains real-time performance while achieving the highest mAP.

Table 2: Comparison with state-of-the-art detection methods.

Method	Backbone	Precision	Recall	mAP@0.5	FPS
Faster R-CNN [Ren et al., 2016]	ResNet-50	91.8	90.2	90.5	12
SSD [Liu et al., 2016]	VGG-16	87.6	85.3	86.1	45
YOLOv5 [Jocher et al., 2022]	CSPDarknet	92.1	90.8	91.3	68
YOLOv7 [Wang et al., 2023]	E-ELAN	93.0	91.5	92.1	72
Proposed YOLOv8n	C2f Backbone	93.5	91.8	92.6	78

4.3 Qualitative Analysis

This section presents sample results from the welding dataset to visually evaluate the proposed method. Figure 2 shows that the YOLOv8n model successfully detects welding defects with accurate bounding boxes.

The model performs well across different conditions, including changes in lighting, weld texture, and background noise. It is also able to correctly identify multiple defect classes such as cracks, porosity, lack of fusion, and spatter. Overall, the qualitative results demonstrate that the model provides reliable and consistent defect detection, with good localization and correct class predictions, even in challenging cases.

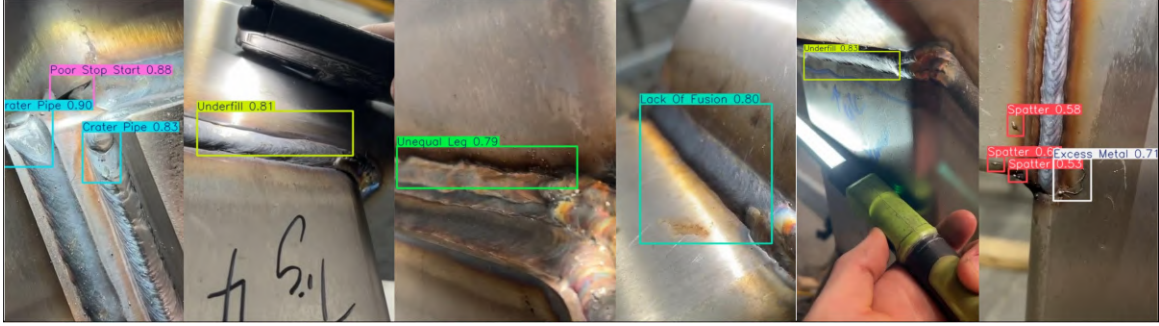


Figure 2: Qualitative analysis of the welding dataset shows promising results for defect detection. The YOLOv8n model performs effectively in identifying and detecting weld defects.

5 Conclusion & Future Work

This work demonstrates effective weld defect detection using a YOLOv8n-based methodology under standard industrial conditions. However, performance may degrade in more challenging scenarios such as low illumination, heavy noise, and partial occlusion. Future work will therefore focus on improving robustness in these conditions and optimizing the model for real-time deployment on edge and resource-constrained industrial devices. In addition, multi-modal fusion with thermal or infrared imaging will be explored to improve defect visibility.

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